

University of Ottawa
Department of Civil Engineering
CVG 6320 – SEISMIC RISK ASSESSMENT & MANAGEMENT OF EXISTING BUILDINGS

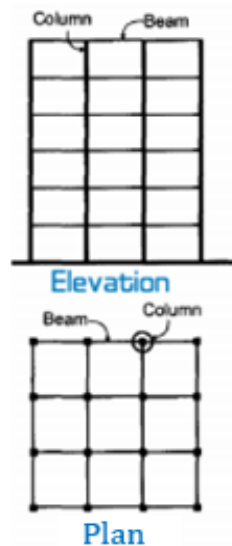
Due Date: Feb. 15th 2025 (at 11:59 PM)

ASSIGNMENT #2

Two six-storey reinforced concrete buildings shown in the figure below were located in Ottawa, ON and were designed according to NBC-1980 (between pre-code and post-benchmark). The building information for these two buildings is provided below:

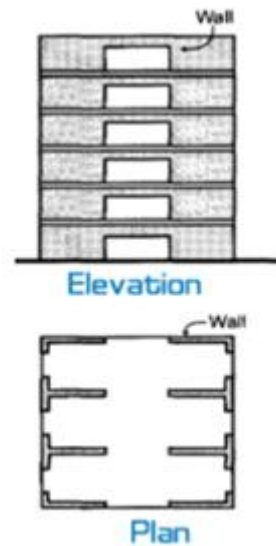
- Storey height = 3.6 m
- SFRS: conventional construction of moment resisting frame or conventional construction of shear wall
- Site class: class C

The floor plans and elevation with the SFRSs for the two buildings are shown in the figure below:



Case 1

conventional construction of moment resisting frames (assume all frames are identical in geometry and stiffness)



Case 2

conventional construction of shear walls

- a) Determine the probabilities of collapse associated with complete damage state for both cases.
- b) Determine the probabilities of collapse associated with complete damage state for both cases if the buildings were constructed in 1915 (Pre-Code).
- c) Determine the probabilities of collapse associated with complete damage state for both cases if the buildings were constructed in 2010 (Post-Benchmark).
- d) Determine the probabilities of collapse associated with complete damage state for both cases if the first storey height is 5.5 m and the storey height of the other storeys is 3.5 m (severe vertical irregularity).

NOTES:

- 1) Use FEMA P-155 methodology to construct the building capacity curve
- 2) Assume the damped demand spectra provided in the table below are valid for part a, b, c and d. (No need to compute the damped demand spectrum)
- 3) Assume the lognormal standard deviation ($\beta_{c,p}$) is 0.75.
- 4) If any factor or coefficient in Table A-1 through A-11 of FEMA P-155 for Pre-Code or Post-Benchmark is not available, use the value for basic score.

Moment Resisting Frame		Shear Wall	
Spectral Acceleration S_a (g)	Spectral Displacement S_d (in)	Spectral Acceleration S_a (g)	Spectral Displacement S_d (in)
0.207	1.559	0.283	1.139
0.203	1.856	0.278	1.356
0.077	1.856	0.105	1.358
0.074	1.902	0.099	1.400
0.070	1.951	0.094	1.450
0.067	2.001	0.088	1.506
0.065	2.055	0.080	1.600
0.062	2.103	0.073	1.706
0.058	2.202	0.065	1.852
0.054	2.302	0.059	2.008
0.051	2.403	0.053	2.204
0.046	2.601	0.048	2.402
0.042	2.801	0.043	2.606
0.039	3.003	0.040	2.800
0.036	3.202	0.037	3.001
0.034	3.352	0.034	3.257